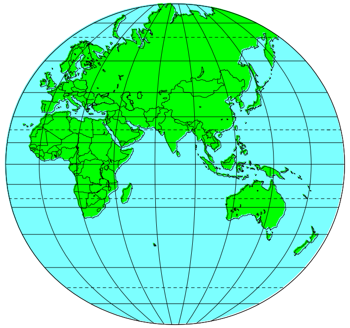
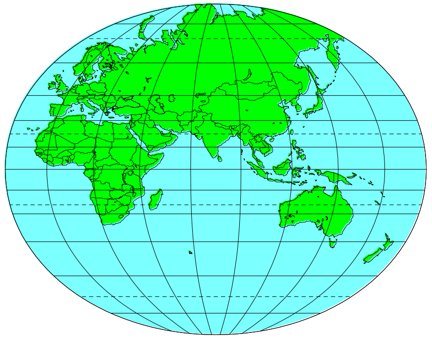
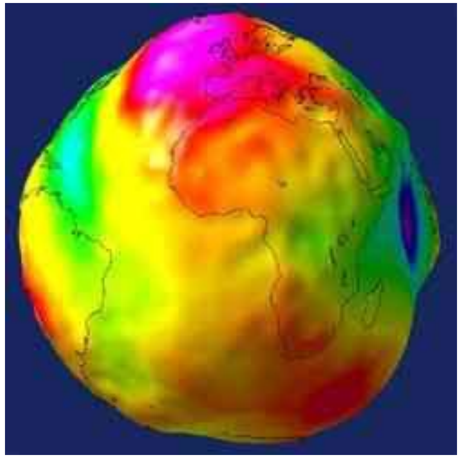
ASEN 5090 Lecture 4
Coordinate Frames

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Week	Date	Topic	Reading	Assignments - Due 11pm*
1	8/24/26	L1. Introduction & GPS Basics	PNT Ch 1, ME 2.1 & 2.2	
	8/26/26	L2. Navigation Measurements & Methods	ME Ch 1	
	8/28/26			HW1 Assigned
2	8/31/26	L3. GPS Signal Structure	ME Ch 2, PNT Ch 2	
	9/2/26	L4. Coordinate Frames	ME Ch 4.1, 4.A	
	9/4/26			HW1 Due, HW2 Assigned
3	9/7/26	Labor Day (no class)		
	9/9/26	L5. Orbits & Visibility	ME Ch 4.3	
	9/11/26			HW3 Assigned

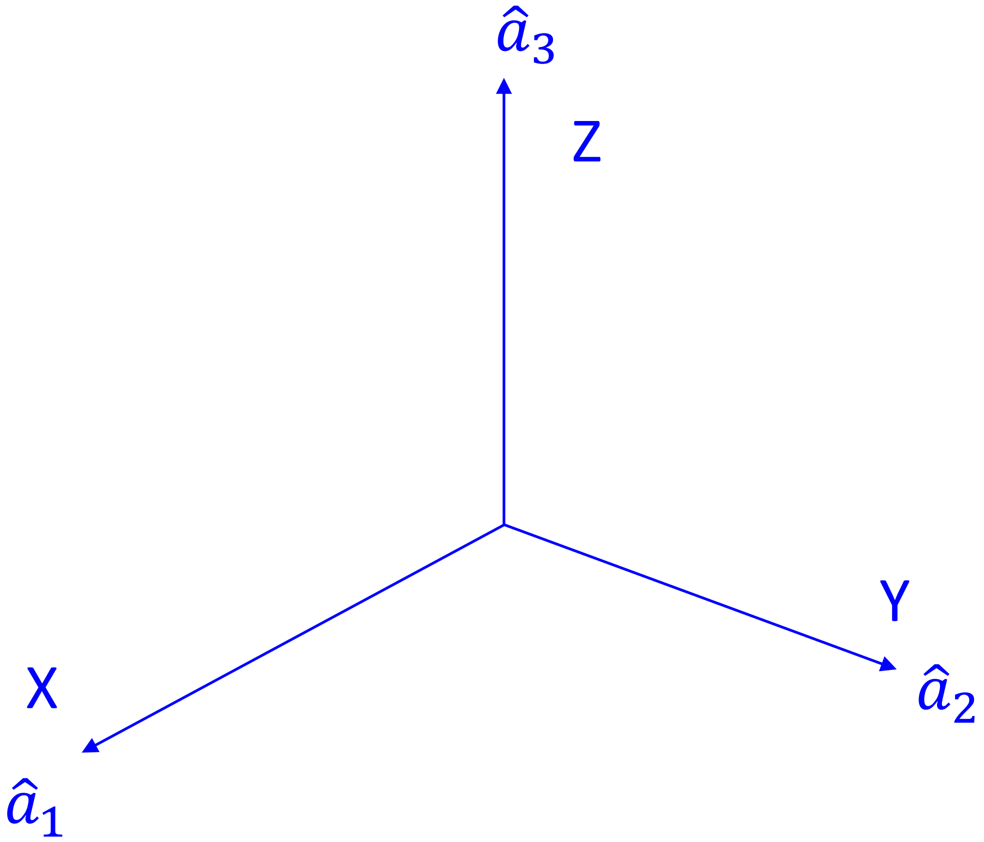
Outline – Coordinate Frames

- Vectors and Transformation matrices
- Earth centered **inertial** (ECI)
- Earth centered Earth fixed (ECEF)
- Transformation between ECI and ECEF
- Latitude, Longitude, Height
- Local Coordinates – East-North-Up (ENU)

- **Reading Assignment**
  - ME Ch 4.1 and 4.A

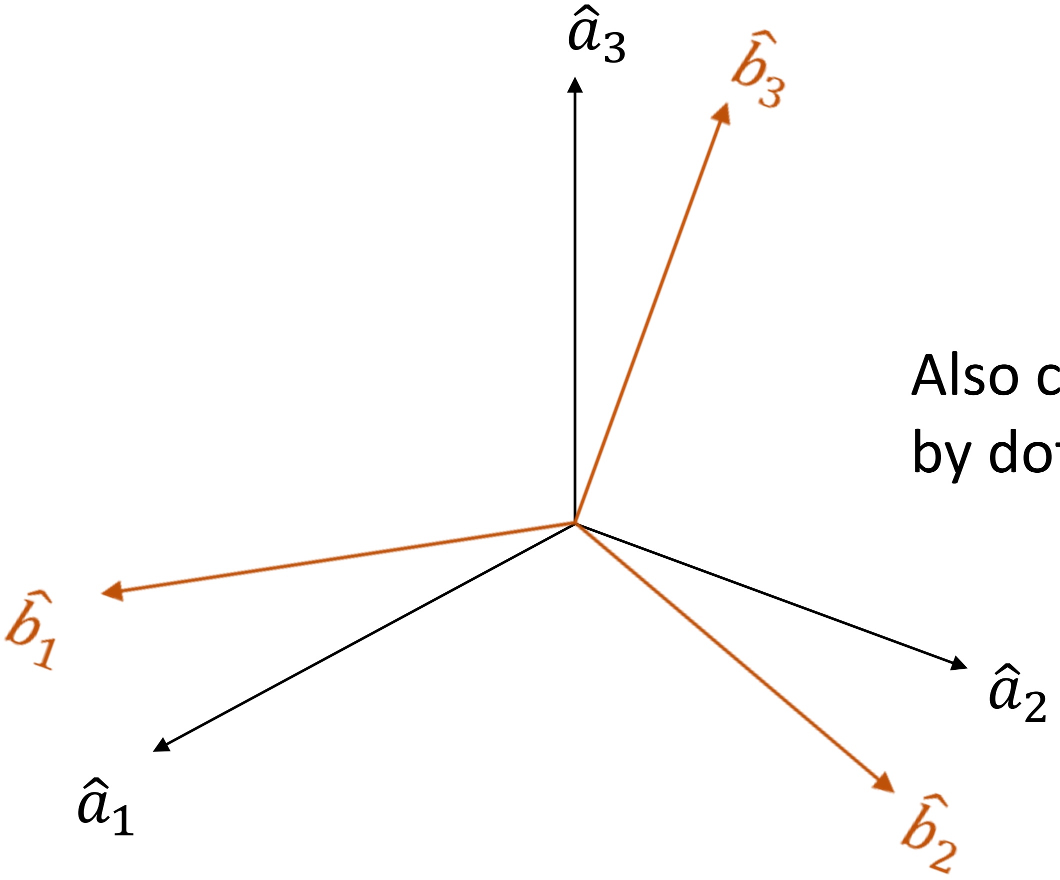
Cartesian Coordinate Frames

- Defined by a center & three orthogonal unit vectors, *i*, *j*, *k*.
- Use a right-handed set : *i* x *j* = *k*



Vectors &
Representation of a Vector in a Frame

Transformation Matrix



The transformation matrix transforms the representation of a vector from one coordinate frame to another

$$\left[\vec{r}\right]_B = C_B^A \left[\vec{r}\right]_A$$

Also called direction cosine matrix (DCM) - can be formed by dot products of unit vectors of two frames

$$C_B^A = \begin{bmatrix} \hat{b}_1 \cdot \hat{a}_1 & \hat{b}_1 \cdot \hat{a}_2 & \hat{b}_1 \cdot \hat{a}_3 \\ \hat{b}_2 \cdot \hat{a}_1 & \hat{b}_2 \cdot \hat{a}_2 & \hat{b}_2 \cdot \hat{a}_3 \\ \hat{b}_3 \cdot \hat{a}_1 & \hat{b}_3 \cdot \hat{a}_2 & \hat{b}_3 \cdot \hat{a}_3 \end{bmatrix}$$

Transformation Matrix Properties

- Nine elements are not independent because it must be orthonormal

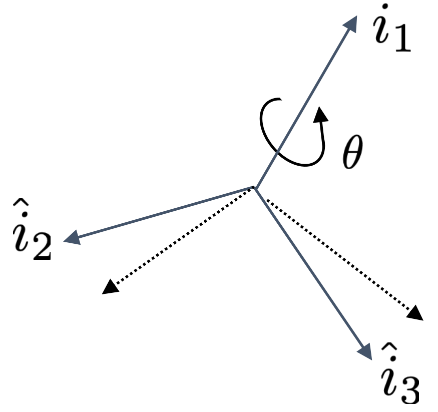
$$C_B^A \, C_A^B = C_A^B \, C_B^A = \mathbf{I}$$

$$C_B^A = \left[C_A^B\right]^T = \left[C_A^B\right]^{-1}$$

TRANSFORMATION MATRICES FOR EULER ROTATIONS

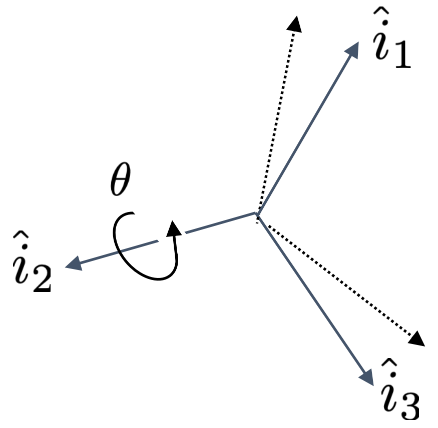
Rotation about 1st axis (e.g. X)

$$C_1(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & s\theta \\ 0 & -s\theta & c\theta \end{bmatrix}$$



Rotation about 2nd axis (e.g. Y)

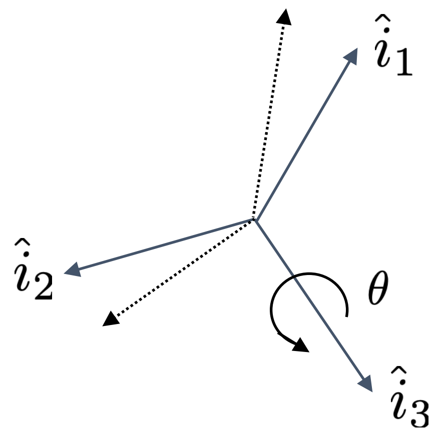
$$C_2(\theta) = \begin{bmatrix} c\theta & 0 & -s\theta \\ 0 & 1 & 0 \\ s\theta & 0 & c\theta \end{bmatrix}$$



TRANSFORMATION MATRICES FOR EULER ROTATIONS

Rotation about 3rd axis (e.g. Z)

$$C_3\left(\theta\right) = \begin{bmatrix} \mathrm{c}\theta & \mathrm{s}\theta & 0 \\ -\mathrm{s}\theta & \mathrm{c}\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



EULER 3-1-3

First $C_3(\phi)$, then $C_1(\theta)$, then $C_3(\psi)$

$$C_{313}\left(\phi,\theta,\psi\right)=\begin{bmatrix}\mathrm{c}\psi & \mathrm{s}\psi & 0\\ -\mathrm{s}\psi & \mathrm{c}\psi & 0\\ 0 & 0 & 1\end{bmatrix}\begin{bmatrix}1 & 0 & 0\\ 0 & \mathrm{c}\theta & \mathrm{s}\theta\\ 0 & -\mathrm{s}\theta & \mathrm{c}\theta\end{bmatrix}\begin{bmatrix}\mathrm{c}\phi & \mathrm{s}\phi & 0\\ -\mathrm{s}\phi & \mathrm{c}\phi & 0\\ 0 & 0 & 1\end{bmatrix}$$

$$C_{313}\left(\phi,\theta,\psi\right)=\begin{bmatrix}\mathrm{c}\psi\mathrm{c}\phi-\mathrm{s}\psi\mathrm{c}\theta\mathrm{s}\phi & \mathrm{c}\psi\mathrm{s}\phi+\mathrm{s}\psi\mathrm{c}\theta\mathrm{c}\phi & \mathrm{s}\psi\mathrm{s}\theta\\ -\mathrm{s}\psi\mathrm{c}\phi-\mathrm{c}\psi\mathrm{c}\theta\mathrm{s}\phi & -\mathrm{s}\psi\mathrm{s}\phi+\mathrm{c}\psi\mathrm{c}\theta\mathrm{c}\phi & \mathrm{c}\psi\mathrm{s}\theta\\ \mathrm{s}\theta\mathrm{s}\phi & -\mathrm{s}\theta\mathrm{c}\phi & \mathrm{c}\theta\end{bmatrix}$$

Practice Problems

Compute the transformation matrix (DCM) for the following sequences, and transform the unit vector [u]' = [1 1 1]/sqrt(3) to each of the new frames.

a) Euler 3-2-1 rotations of 35 deg each
b) Euler 3-1-3 rotations of 35 deg each
c) Euler 3-2-1 rotations of 10 deg, 20 deg, -30 deg, respectively
d) Euler 1-2-3 rotations of 10 deg, 20 deg, -30 deg, respectively

ANSWERS

u =

0.5774
0.5774
0.5774

Ca =

0.6710	0.4698	-0.5736
-0.2004	0.8597	0.4698
0.7139	-0.2004	0.6710

ua =

0.3275
0.6519
0.6839

Cb =

0.4015	0.8547	0.3290
-0.8547	0.2207	0.4698
0.3290	-0.4698	0.8192

ub =

0.9152
-0.0948
0.3916

Cc =

0.9254	0.1632	-0.3420
-0.3188	0.8232	-0.4698
0.2049	0.5438	0.8138

uc =

0.4310
0.0199
0.9021

Cd =

0.8138	-0.4410	-0.3785
0.4698	0.8826	-0.0180
0.3420	-0.1632	0.9254

ud =

-0.0033
0.7704
0.6375

ECI & ECEF

Used for descriptions of satellite motion*:**

Earth Centered Inertial (ECI)

Conventional Inertial Reference System (CIRS)

Geocentric Celestial Reference Frame (GCRF)

Used for descriptions of ground-based things:

Earth Centered Earth Fixed (ECEF)

Conventional Terrestrial Reference System (CTRS)

International Terrestrial Reference Frame (ITRF)

World Geodetic System (WGS 84)

*** GPS satellite positions are given in WGS-84

Earth Centered Inertial (ECI)

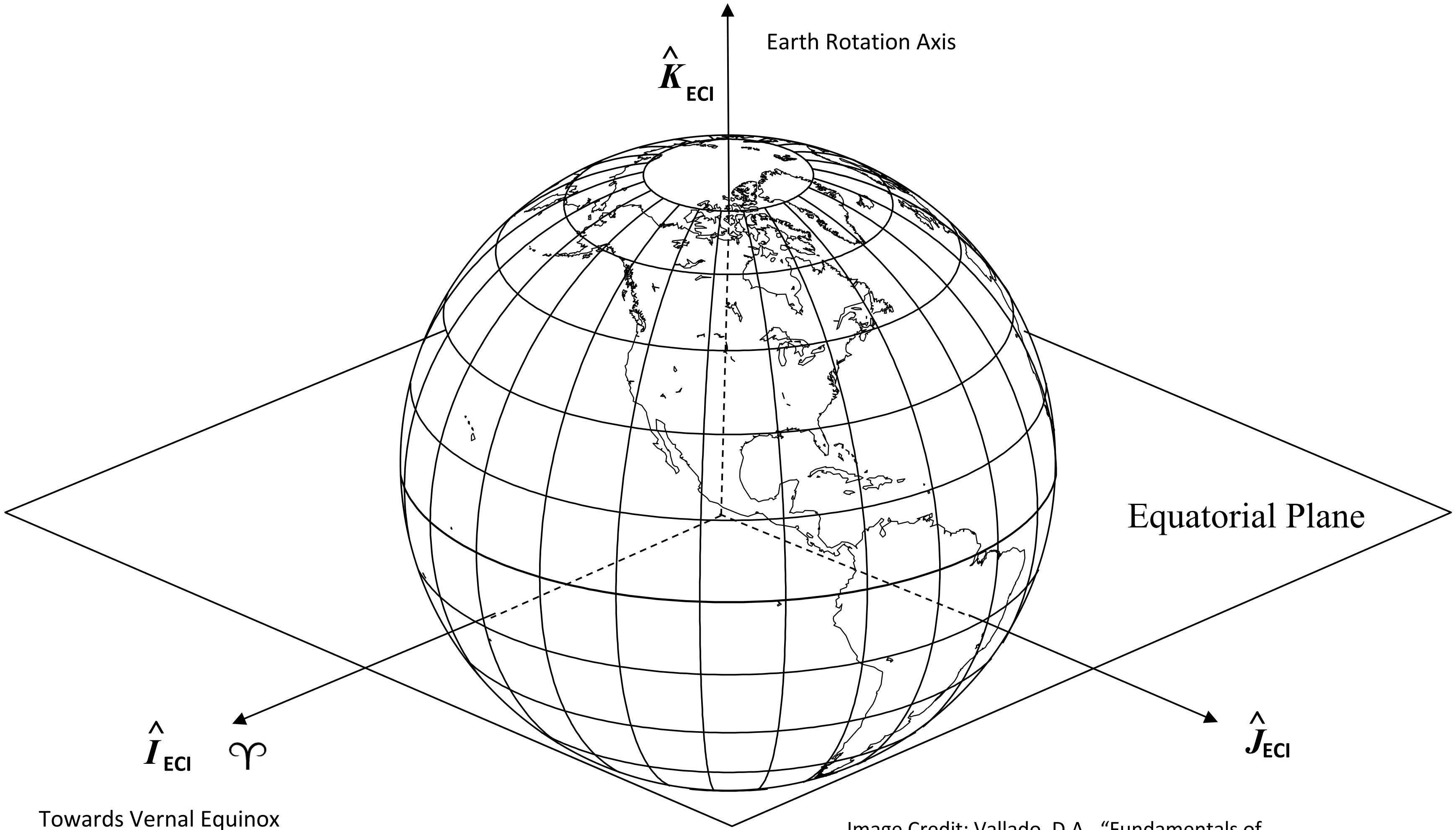
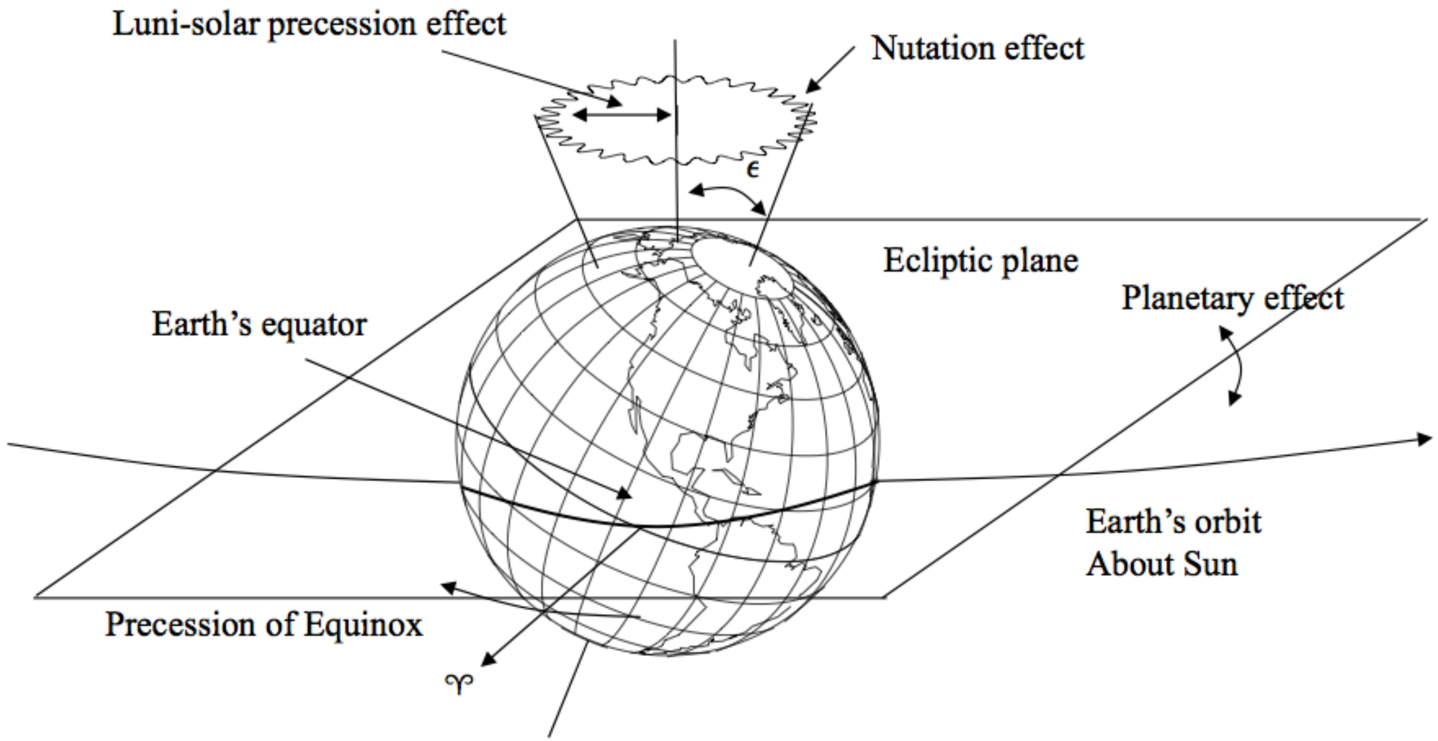


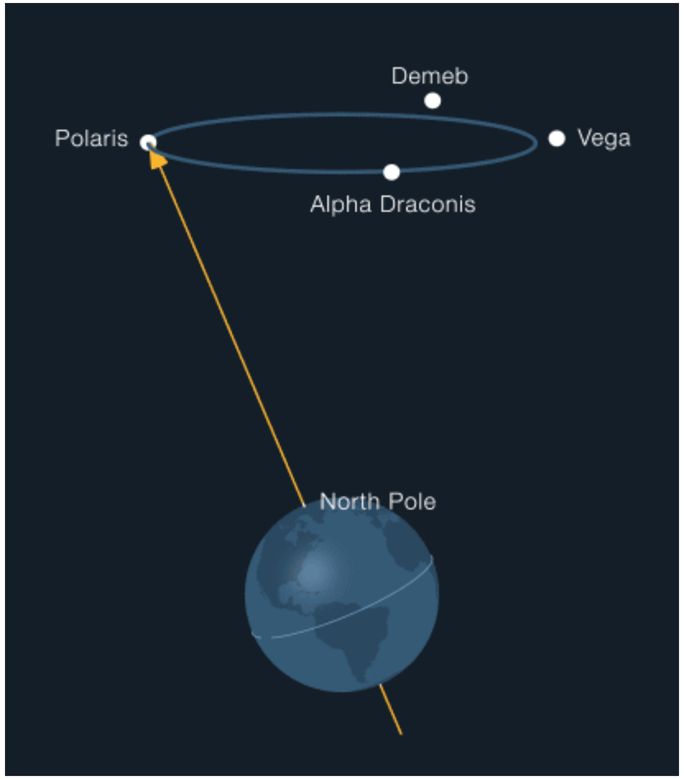
Image Credit: Vallado, D.A., "Fundamentals of Astrodynamics and Applications," 3rd ed., 2007, pp. 159.

- Origin at earth center of mass
- X – defined by direction of the vernal equinox (through the true equator)
- Z – defined by axis of rotation of the Earth
- Y – right-handed set (also through true equator)

- Not exactly inertial because origin accelerates as Earth travels around the sun, and the rotation axis precesses and nutates
- The term ECI is a generic term for this system.
- The International Earth Rotation Service (IERS) provides a precise definition which is called the Geocentric Celestial Reference Frame (GCRF) (*see Vallado*)

Precession/Nutation

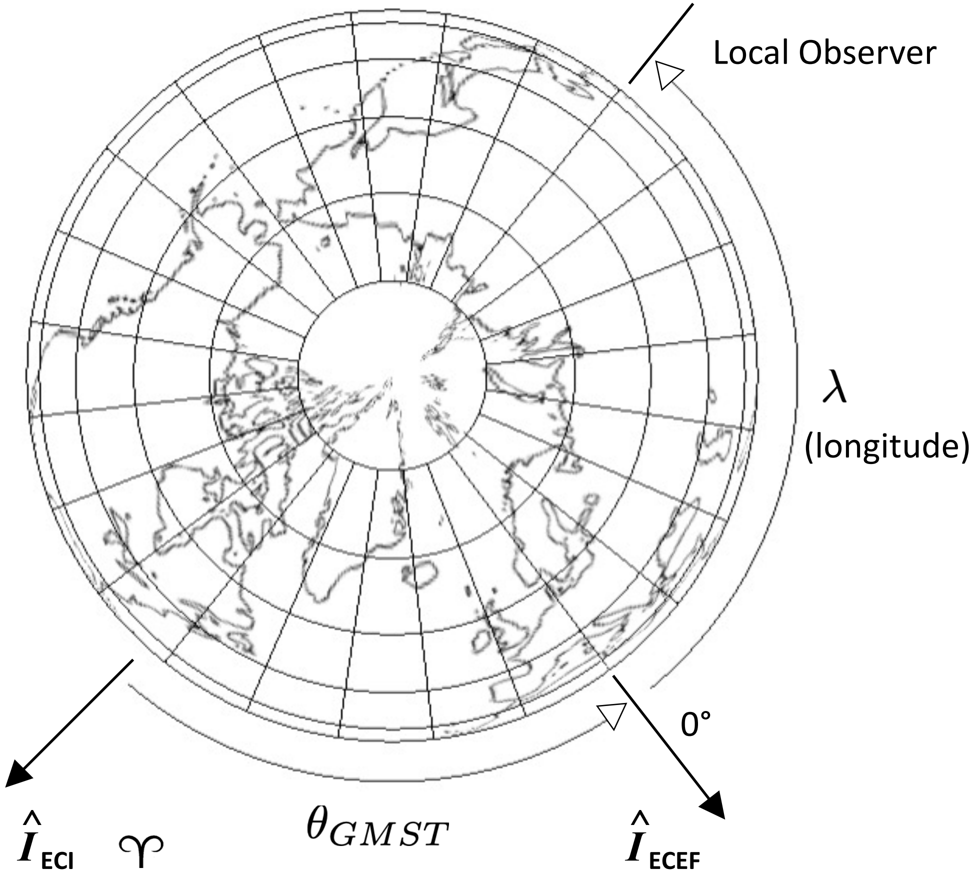




- Precession is ~23.5 deg cone and is caused by the gravitational attraction of the Sun and Moon on the non-spherical Earth (period 26,000 yrs)
- Nutation is a small bobbing motion with period of 18.6 years (due to moon)
- A spherical Earth with homogeneous density distribution would neither precess nor nutate

https://solarsystem.nasa.gov/basics/chapter2-1/#:~:text=This%20precession%20causes%20the%20axis,around%20before%20it%20falls%20over.

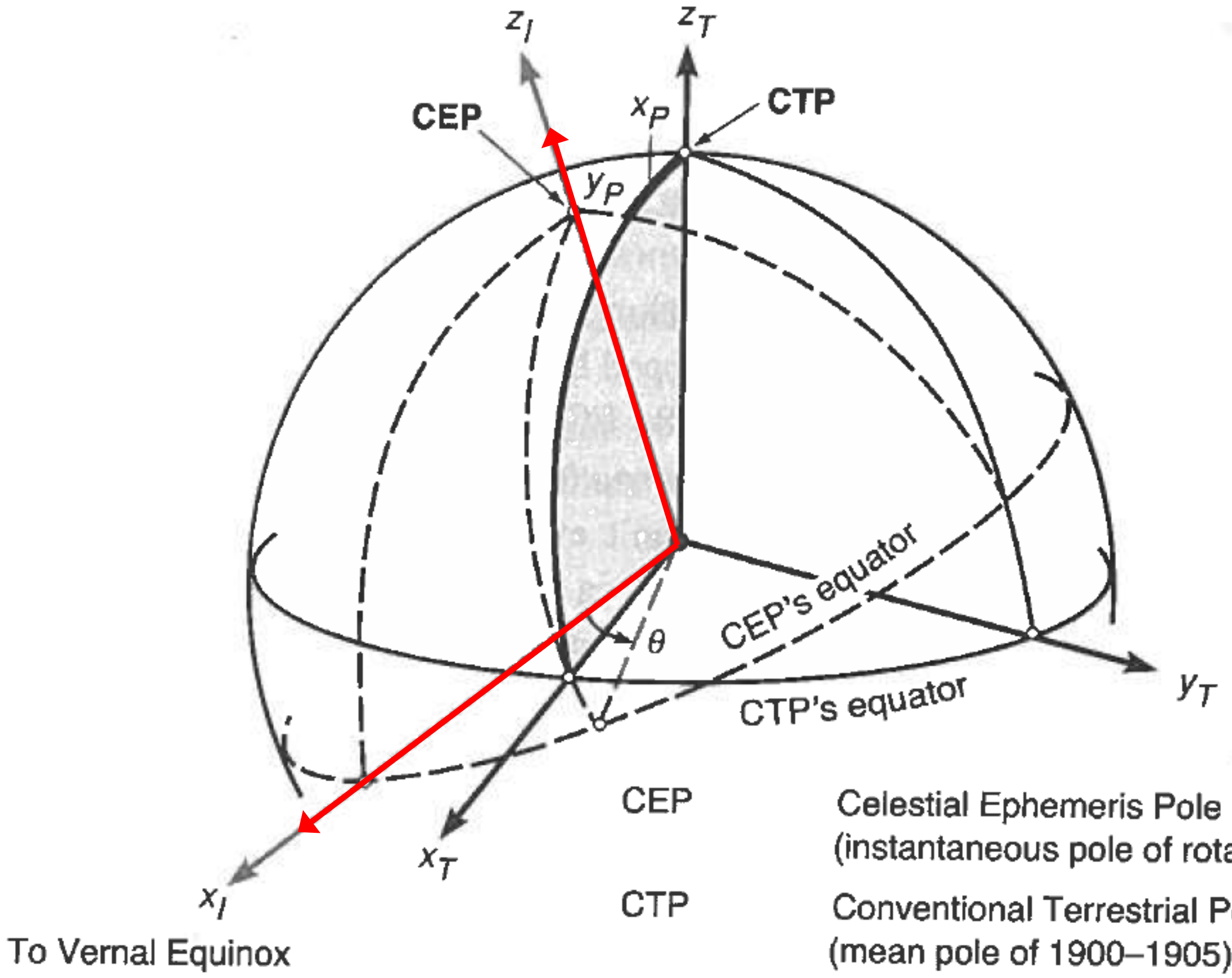
EARTH CENTERED EARTH FIXED (ECEF)



- Origin at earth center of mass
- International Terrestrial Reference Frame (ITRF) is the formal terminology
  - Z - axis through IRP – average position of rotation axis between 1900 and 1905 (north pole)
  - X - equatorial through a reference meridian (Greenwich meridian, 0° longitude)
  - Y - make it right-handed (90 degrees East of X)
- Realization is based on a selection of a globally distributed set of points.
- WGS 84 – World Geodetic System 1984 is one of these realizations
- Offset between ECI and ECEF (simplified)
  - θ angle between vernal equinox and Greenwich meridian is Greenwich Apparent Sidereal Time, GAST (a simplification is to use Greenwich Mean Sidereal Time, GMST)
  - Polar motion
    - the instantaneous axis of rotation moves around wrt the Earth's crust (polar motion)
    - xp, yp, represent the offset of the instantaneous axis of rotation (the Celestial Intermediate Pole) from the International Reference Pole (a fixed point on the Earth surface that serves as Z, aka the north pole)

Relationship Between CIRF & CTRF (ECEF)

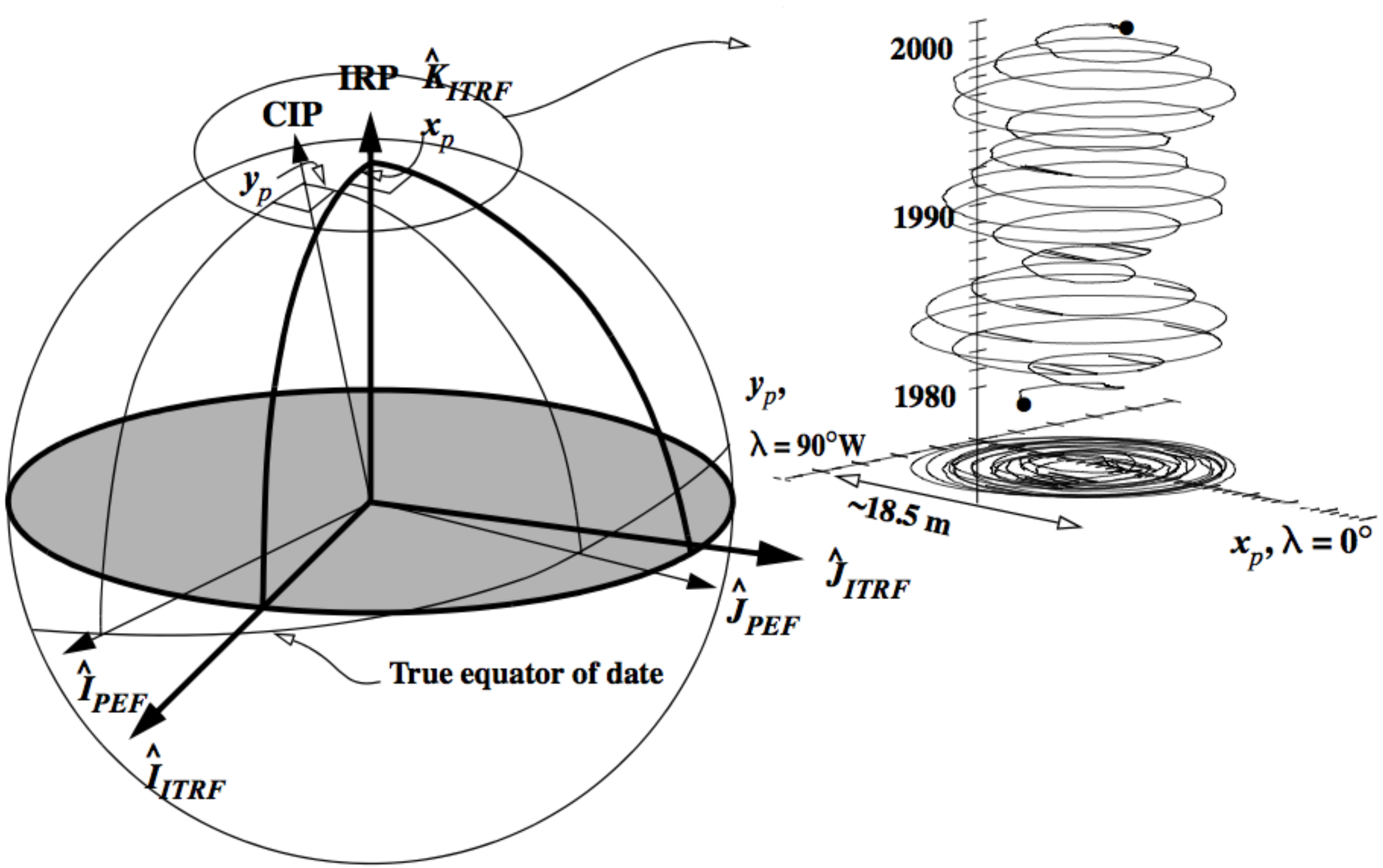
Celestial = Inertial
Terrestrial = Earth Fixed



CEP	Celestial Ephemeris Pole (instantaneous pole of rotation)
CTP	Conventional Terrestrial Pole (mean pole of 1900–1905)
(x_p, y_p)	Polar motion coordinates (in CTRF)
θ	Greenwich Apparent Sidereal Time (GAST)

Textbook, P96, Fig. 42.

Polar Motion – Earth Orientation Parameters (EOPs)



Similar image in Misra and Enge Ch 4

- **CIP:** *Celestial Intermediate Pole* (instantaneous rotation axis of the Earth)
- **IRP:** *International Reference* Pole (agreed upon "north pole" of Earth for ITRF)
- $\boldsymbol{x_p, y_p}$: polar motion coordinates
- Polar motion cannot be predicted. x_p and y_p are observed parameters

Transformation from ECI to ECEF

Full Precise Transformation

$$[\mathbf{r}]_{ECEF} = [W(t)]\,[R(t)]\,[N(t)]\,[P(t)]\,[\mathbf{r}]_{ECI}$$

Polar Motion

Sidereal-Rotation

Nutation

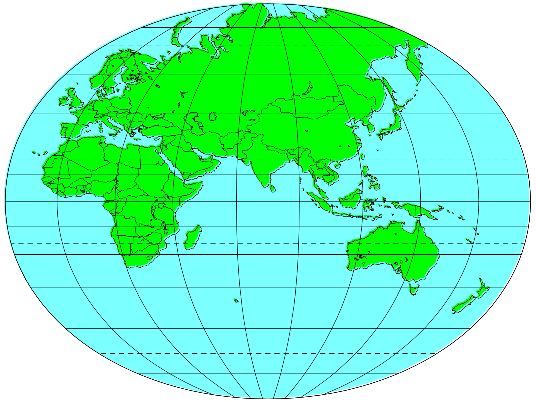
Precession

Simplified Transformation

$$[\mathbf{r}]_{ECEF} \approx \begin{bmatrix} cos(\theta_{GMST}) & sin(\theta_{GMST}) & 0 \\ -sin(\theta_{GMST}) & cos(\theta_{GMST}) & 0 \\ 0 & 0 & 1 \end{bmatrix} [\mathbf{r}]_{ECI}$$

- Use rotation about z-axis through Greenwich Mean Sidereal Time (GMST) only
- Accurate to about 100 meters in LEO and 1 km at GEO

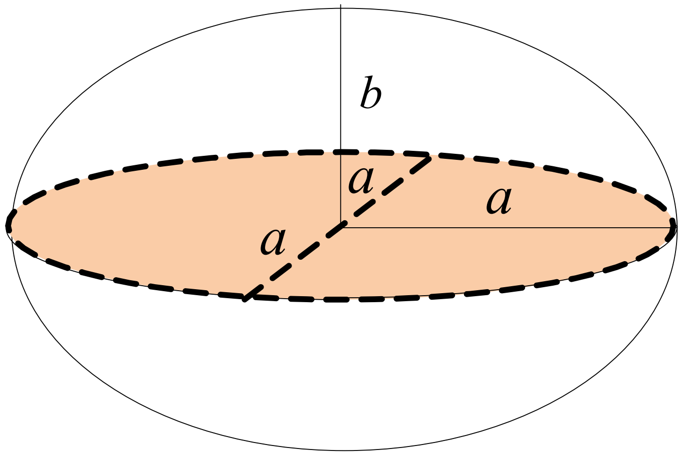
Geodetic Coordinate – Latitude, Longitude, Height



- Angular coordinates are convenient/intuitive
  Requires a smooth model for the earth
- Geocentric latitude goes through the center of the ellipse
- Shape of the Earth is approximately an ellipsoid – fatter at the equator, flattened at the poles.
  - Parallels (latitudes) are circles. Meridians (longitudes) are ellipses.
- Geodetic coordinates are referenced to this ellipsoid:
  - Geodetic latitude - angle between the equatorial plane and a normal to the ellipsoid surface at P.
  - Geodetic longitude angle between meridian and reference
  - Geodetic or ellipsoidal height is distance along a line normal to the ellipsoid at P (Figure 4.4)
- The ellipsoid is just a mathematical figure or reference surface.
- Countries/Regions pick an ellipsoid that suits them - so the “height” wrt these local reference frames can change at international boundaries.

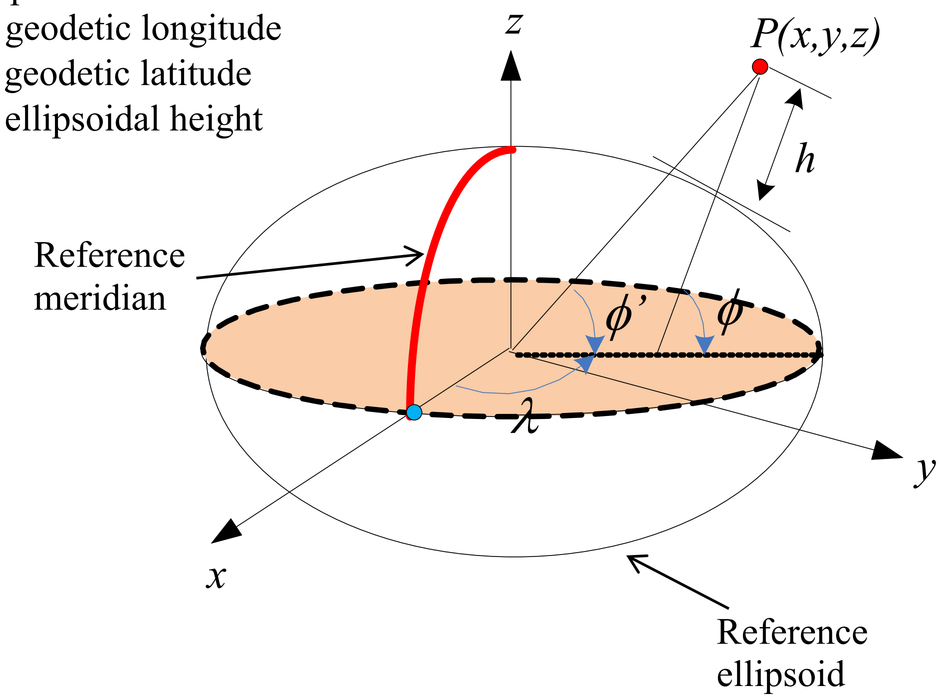
Ellipsoid and Ellipsoidal Coordinates

Oblate ellipsoid
a: semi-major axis
b: semi-minor axis
e: eccentricity
f: flattening



$$e^2 = \frac{a^2 - b^2}{a^2} \qquad f = \frac{a-b}{a}$$

Ellipsoidal coordinates
λ : geodetic longitude
ϕ : geodetic latitude
h : ellipsoidal height



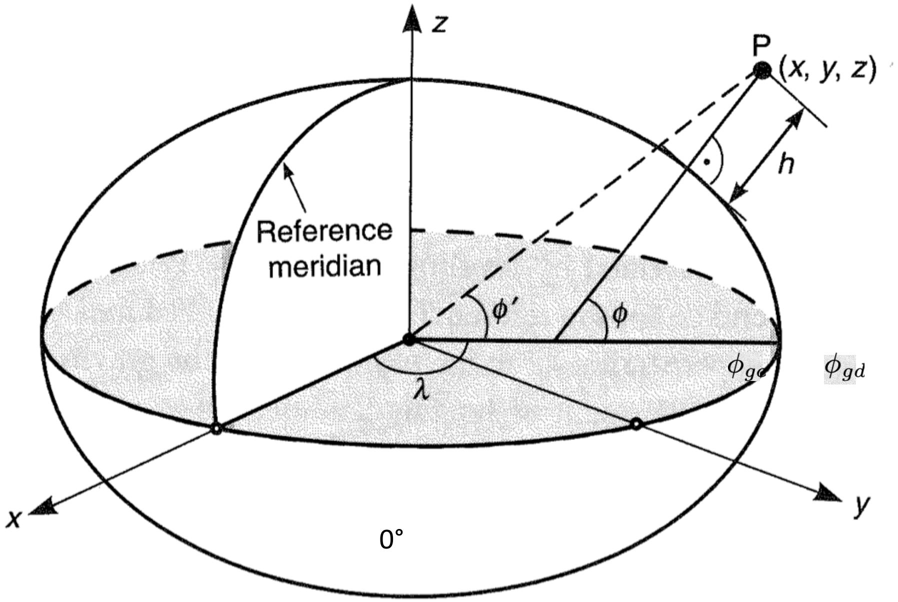
World Geodetic System 1984 (WGS 84)

- What does it do?
  - Defines the ECEF Cartesian coordinate frame
  - Defines the Earth ellipsoid
  - Models the Earth gravity field
- History
  - Defense Mapping Agency (DMA) defined CTRS, 1984
  - Became part of National Imagery and Mapping Agency (NIMA), 1996
  - National Geospatial Intelligence Agency (NGA), 2004
  - Official agency for mapping, charting, navigation, & geodetic products for DoD
  - Essential to GPS development
  - The use of GPS enhanced WGS 84 as a world standard

WGS 84 Fundamental Parameters (Revised 1997)

Parameters	Value
Ellipsoid semi-major axis (a)	6378137.0 m
Ellipsoid reciprocal flattening (1/f)	298.257223563
Earth angular velocity (ω_E)	$7.2921151467 \times 10^{-5}$ rad/sec
Geocentric gravitational constant (GM)	$3.986004418 \times 10^{14}$ m^3/sec^2

Geodetic (Lat/Lon/Ellipsoidal Height) to ECEF



$$\lambda = \text{longitude}$$
$$\phi_{gd} = \text{geodetic latitude}$$
$$\phi_{gc} = \text{geocentric latitude}$$

Image from Misra and Enge Ch 4

$$C_\oplus = \frac{R_\oplus}{(1 - e_\oplus^2 \sin^2 \phi_{gd})^{1/2}}$$

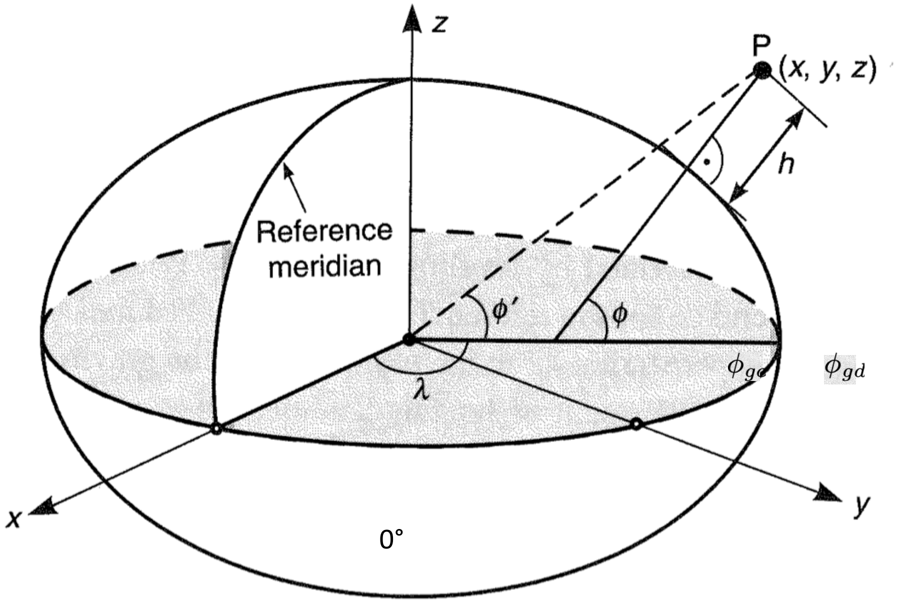
$$\begin{bmatrix} x \\ y \\ z \end{bmatrix}_{ECEF} = \begin{bmatrix} (C_\oplus + h)\cos\phi_{gd}\cos\lambda \\ (C_\oplus + h)\cos\phi_{gd}\sin\lambda \\ (C_\oplus(1 - e_\oplus^2) + h)\sin\phi_{gd} \end{bmatrix}$$

WGS-84 Ellipsoid Parameters

$R_\oplus = 6378.137\ \text{km}$	(Semi-major axis of ellipsoid)
$f = 1/298.257223563$	(flattening parameter of ellipsoid)
$e_\oplus^2 = 2f - f^2$	(square of eccentricity of ellipsoid)

Note: $C_\oplus$ is known as the *radius of curvature in the meridian*

ECEF to Geodetic (Lat/Lon/Ellipsoidal Height)



$$\lambda = \text{longitude}$$
$$\phi_{gd} = \text{geodetic latitude}$$
$$\phi_{gc} = \text{geocentric latitude}$$

Image from Misra and Enge Ch 4

$$\tan\lambda = y/x$$

$$\rho = \sqrt{x^2 + y^2}$$

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\text{Initial Guess: } \phi_{gd} = \sin^{-1}\left(\frac{z}{r}\right)$$

LOOP

$$C_\oplus = \frac{R_\oplus}{(1 - e_\oplus^2 \sin^2\phi_{gd})^{1/2}}$$

$$\tan\phi_{gd} = \frac{z + C_\oplus e_\oplus^2 \sin\phi_{gd}}{\rho}$$

UNTIL $\phi_{gd} - \phi_{gd_{old}} < \textit{tolerance}$ (~1E-8)

$$h = \frac{\rho}{\cos\phi_{gd}} - C_\oplus$$

Heights

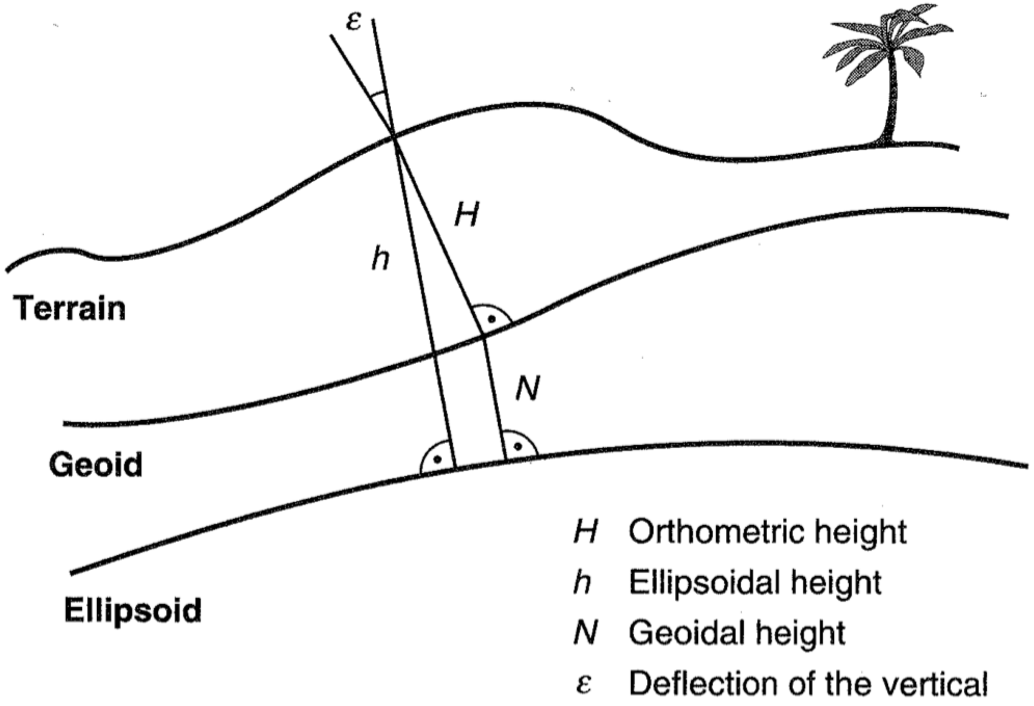


Image Credit: Misra and Enge Ch 4

$$H \approx h - N$$

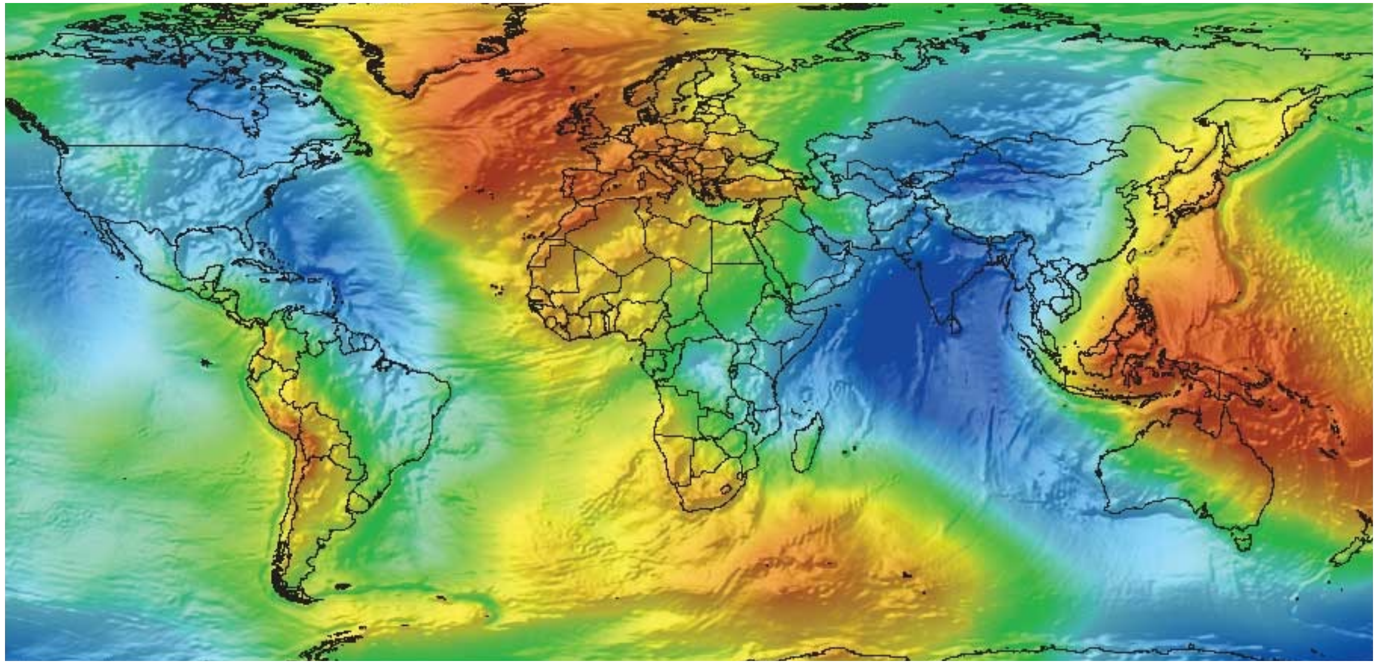
Ellipsoidal Height (h): height of object wrt reference ellipsoid (e.g. WGS-84 ellipsoid, used by GPS)

Orthometric Height (H): height of object wrt reference geoid (e.g. WGS-84 geoid or EGM-96 geoid)
- orthometric height is commonly known as height above mean sea level

GEOID

- The GEOID is a surface where gravity is constant (aka Mean Sea Level)
- Examples: WGS-84 geoid, EGM-96 geoid (also wrt WGS-84 ellipsoid)
- Typically represented using spherical harmonics
- Geoid height varies from -104m to +75m, RMS is 30m worldwide (represents how well the reference ellipsoid fits the geoid)

EGM96 geoid undulations wrt WGS84 ellipsoid

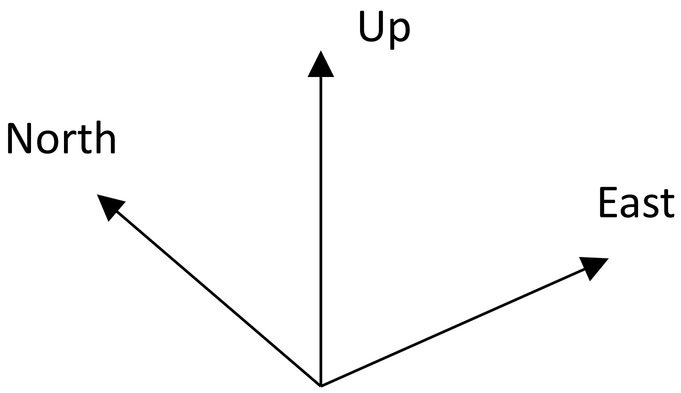


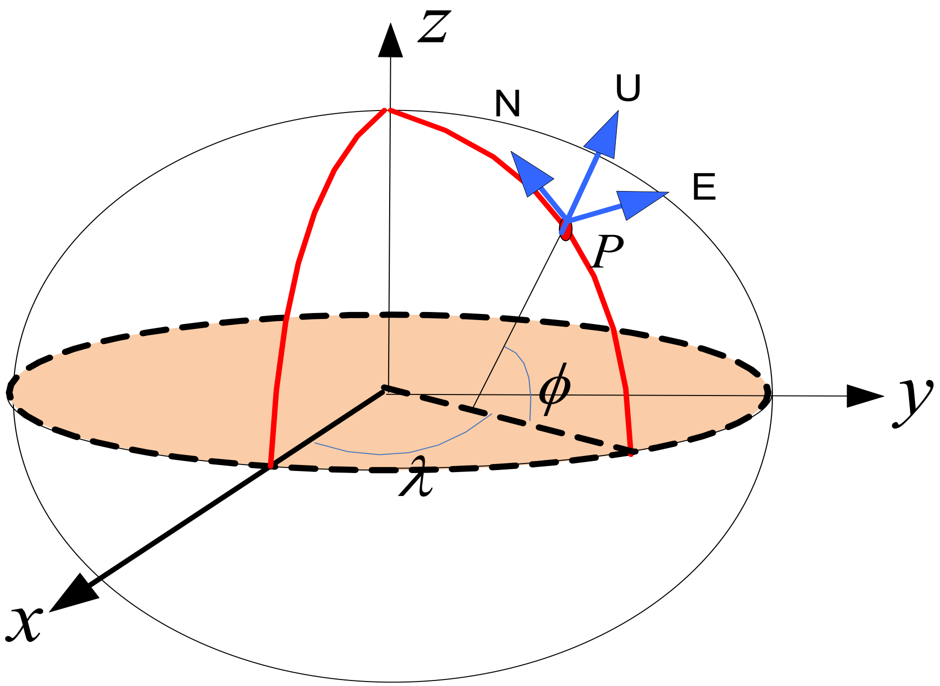
Geoid Height Calculator:
https://www.unavco.org/software/geodetic-utilities/geoid-height-calculator/geoid-height-calculator.html

MATLAB built-in function (Aerospace Toolbox):
N = geoidheight(latitude,longitude,modelname)

What is ENU?

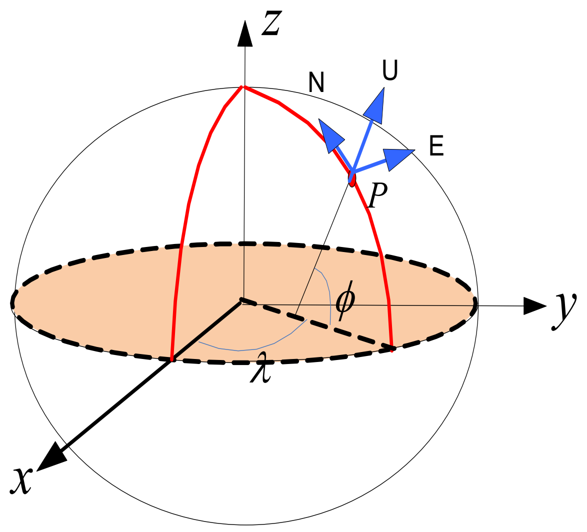
- Local Topocentric Coordinate System





- East-North plane is tangent to surface of the Earth at specified point

Rotation from ECEF to ENU



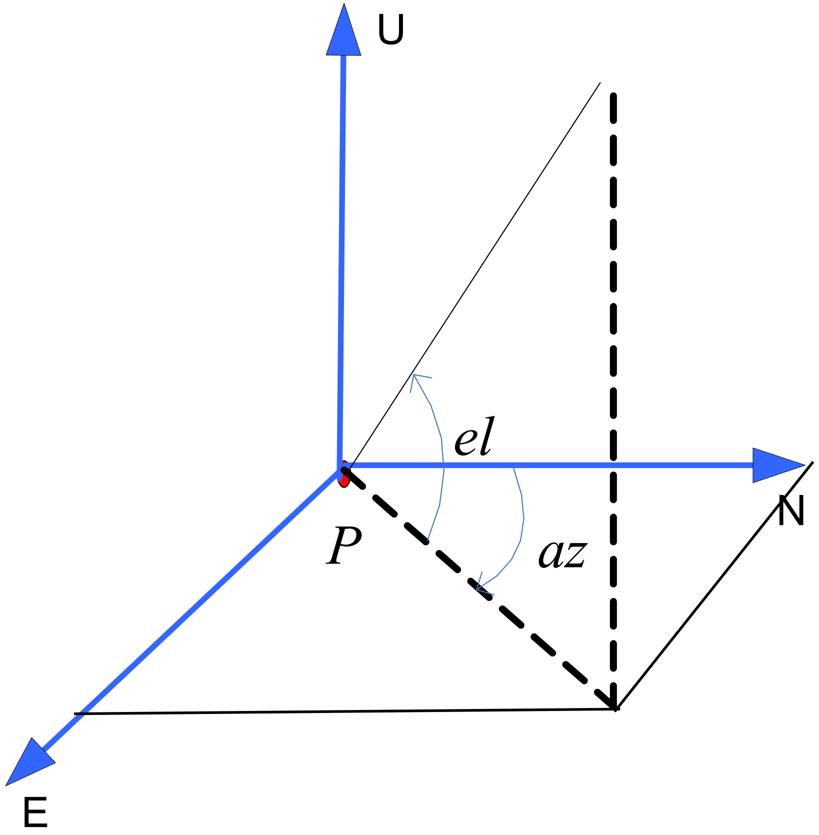
- First $C_3(\lambda+90^o)$ then $C_1(90^o-\phi)$

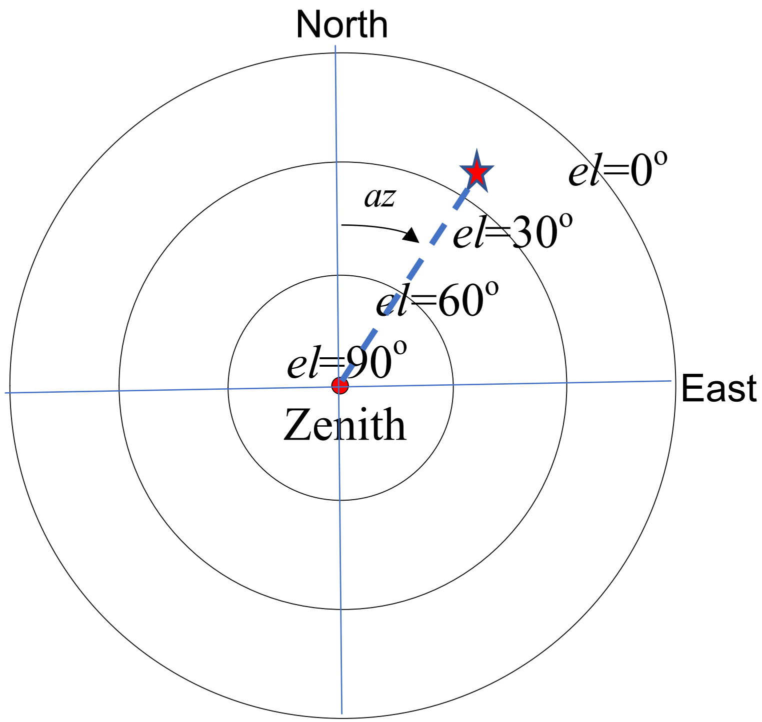
$$C_{ENU}^{ECEF} = C_1(90° - \phi)C_3(90° + \lambda)$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(90^\circ-\phi) & \sin(90^\circ-\phi) \\ 0 & -\sin(90^\circ-\phi) & \cos(90^\circ-\phi) \end{bmatrix} \begin{bmatrix} \cos(90^\circ+\lambda) & \sin(90^\circ+\lambda) & 0 \\ -\sin(90^\circ+\lambda) & \cos(90^\circ+\lambda) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & \sin\phi & \cos\phi \\ 0 & -\cos\phi & \sin\phi \end{bmatrix} \begin{bmatrix} -\sin\lambda & \cos\lambda & 0 \\ -\cos\lambda & -\sin\lambda & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -\sin\lambda & \cos\lambda & 0 \\ -\sin\phi\cos\lambda & -\sin\phi\sin\lambda & \cos\phi \\ \cos\phi\cos\lambda & \cos\phi\sin\lambda & \sin\phi \end{bmatrix}$$

Satellite Elevation and Azimuth





Sky map

$$\tan az = \frac{x_E}{x_N} \qquad \sin el = \frac{x_U}{\sqrt{x_E^2 + x_N^2 + x_U^2}}$$

Acknowledgements

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NEXT TIME – ORBITS